SQL & Tableau

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1 Code & Graphs

This was a quick project to familiarize myself with SQL and Tableau. I started by finding a credit risk data set on Kaggle from UCI machine learning. This data set ended up having a missing variable called risk. I had to find the original dataset and download that instead link. The data had 20 different attribute variables labeled A1, A2, etc. I uploaded this data set to SQL and renamed each of the column names.

```
-- Import our dataset
SELECT *
FROM [dbo].[german_credit_data];

-- Rename the attribute column names to match the german.doc
     atrribute names
EXEC sp_rename 'dbo.german_credit_data.Attribute1', '
    checking_account', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute2', '
    duration', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute3', '
    credit_history', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute4', 'purpose
    ', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute5', '
    credit_amount', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute6', 'savings
    ', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute7', '
    employment_since', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute8', '
    installment_rate', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute9', '
    personal_status_sex', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute10', '
    other_debtors', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute11', '
    residence_since', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute12', '
    property', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute13', 'age',
    'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute14', '
    other_installment_plans', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute15', '
    housing', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute16', '
    existing_credits', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute17', 'job',
    'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute18', '
```

```
    dependents', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute19', '
    telephone', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.Attribute20', '
    foreign_worker', 'COLUMN';
EXEC sp_rename 'dbo.german_credit_data.[class]', '
    credit_risk', 'COLUMN';
```

Followed by a check to see if it was done correctly:

```
-- Check to see if it was done succesfully
SELECT COLUMN_NAME,
       ORDINAL_POSITION,
       DATA_TYPE
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_SCHEMA = 'dbo'
  AND TABLE_NAME = 'german_credit_data'
ORDER BY ORDINAL_POSITION;
```

Then I looked at credit risk as a frequency to create a histogram in Tableau:

```
-- Credit Risk Frequency
SELECT credit_risk,
       COUNT(*) AS frequency
FROM dbo.german_credit_data
GROUP BY credit_risk;
```

This was then followed by credit risk by age:

```
-- Credit risk by age
SELECT age,
       credit_risk,
       COUNT(*) AS frequency
FROM dbo.german_credit_data
GROUP BY age, credit_risk
ORDER BY age, credit_risk;
```

Then the credit amount by age group and credit risk. I created age demographic bin in SQL that way there would be less work to do in Tableau.

```
-- Credit amount by age group and credit risk
SELECT
    CASE
        WHEN age BETWEEN 16 AND 24 THEN '16-24'
        WHEN age BETWEEN 25 AND 34 THEN '25-34'
        WHEN age BETWEEN 35 AND 44 THEN '35-44'
        WHEN age BETWEEN 45 AND 54 THEN '45-54'
        WHEN age BETWEEN 55 AND 64 THEN '55-64'
        WHEN age >= 65 THEN '65+'
        ELSE 'Outside range'
    END AS age_group,
    credit_amount,
```

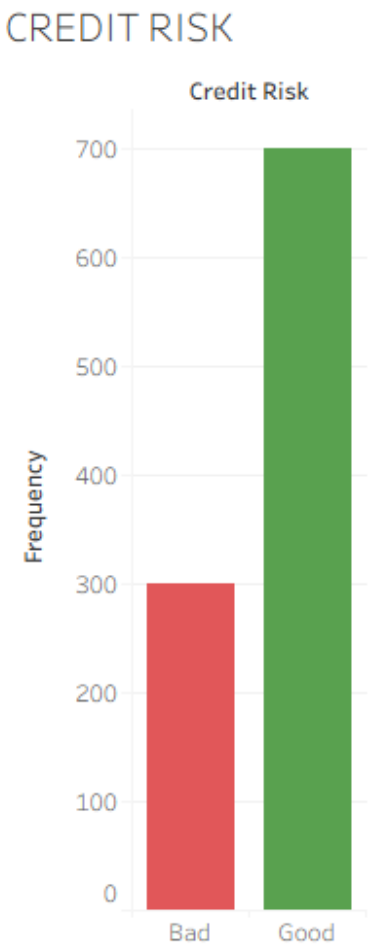


Figure 1: Credit Risk

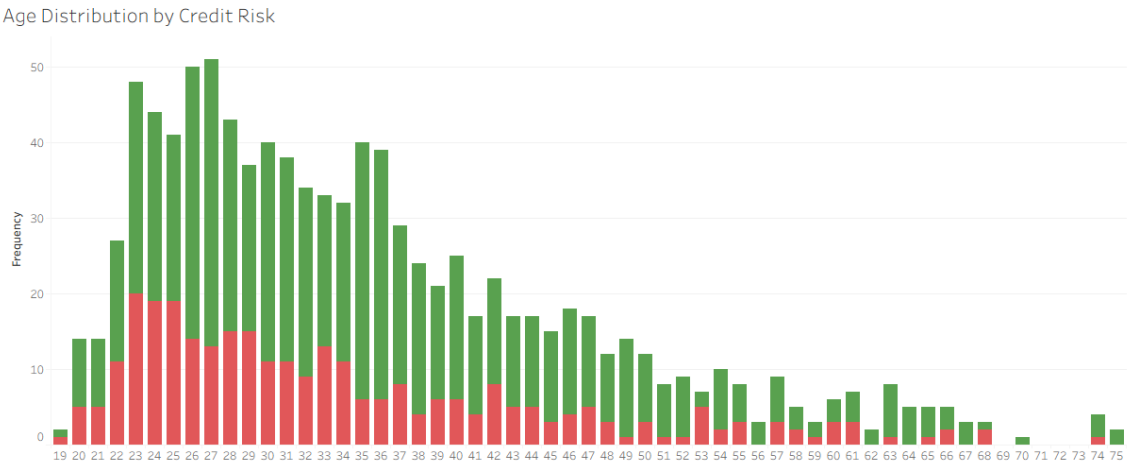


Figure 2: Credit Risk by Age

```
    credit_risk,
    COUNT(*) AS frequency
FROM dbo.german_credit_data
GROUP BY
    CASE
        WHEN age BETWEEN 16 AND 24 THEN '16-24'
        WHEN age BETWEEN 25 AND 34 THEN '25-34'
        WHEN age BETWEEN 35 AND 44 THEN '35-44'
        WHEN age BETWEEN 45 AND 54 THEN '45-54'
        WHEN age BETWEEN 55 AND 64 THEN '55-64'
        WHEN age >= 65 THEN '65+'
        ELSE 'Outside range'
    END,
    credit_amount,
    credit_risk
ORDER BY MIN(age), credit_amount, credit_risk;
```



Figure 3: Credit Amount By Age & Risk